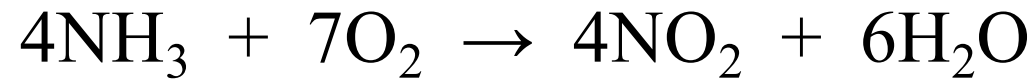


Chapter 14

Reaction Mechanisms

1. Which one of the following is not a valid expression for the rate of the reaction below? E



A) $-\frac{1}{7} \frac{\Delta[\text{O}_2]}{\Delta t}$

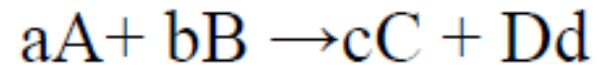
B) $\frac{1}{4} \frac{\Delta[\text{NO}_2]}{\Delta t}$

C) $\frac{1}{6} \frac{\Delta[\text{H}_2\text{O}]}{\Delta t}$

D) $-\frac{1}{4} \frac{\Delta[\text{NH}_3]}{\Delta t}$

E) All of the above are valid expressions of the reaction rate.

Stoichiometry.



$$\text{rate} = -\frac{1}{a} \frac{\Delta[\text{A}]}{\Delta t} = -\frac{1}{b} \frac{\Delta[\text{B}]}{\Delta t} = \frac{1}{c} \frac{\Delta[\text{C}]}{\Delta t} = \frac{1}{d} \frac{\Delta[\text{D}]}{\Delta t}$$

2. The data in the table below were obtained for the reaction:



The rate law for this reaction is rate = E .

A) $k[\text{A}][\text{B}]$

B) $k[\text{P}]$

C) $k[\text{A}]^2[\text{B}]$

D) $k[\text{A}]^2[\text{B}]^2$

E) $k[\text{A}]^2$

Experiment Number	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.273	0.763	2.83
2	0.273	1.526	2.83
3	0.819	0.763	25.47

x3

Not change

x9

3. The data in the table below were obtained for the reaction:



The magnitude of the rate constant is A .

- A) 38.0
- B) 0.278
- C) 13.2
- D) 42.0
- E) 2.21

Experiment Number	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.273	0.763	2.83
2	0.273	1.526	2.83
3	0.819	0.763	25.47

$$\text{rate} = k[A]^m[B]^n$$

$$K = 25.47 / (0.819)^2$$

4. The reaction: $\text{NO}_2 \rightarrow \text{NO} + \frac{1}{2} \text{O}_2$ follows second-order kinetics. At 300 °C, $[\text{NO}_2]$ drops from 0.0100 M to 0.00650 M in 100.0 s. The rate constant for the reaction is E $\text{M}^{-1}\text{s}^{-1}$.

A) 0.096

B) 0.65

C) 0.81

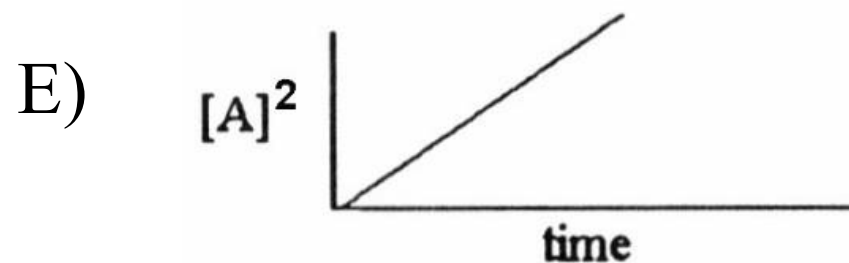
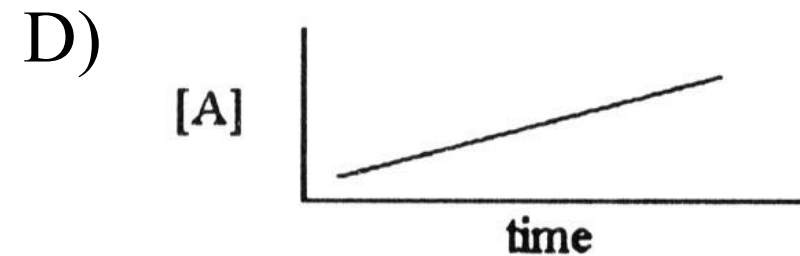
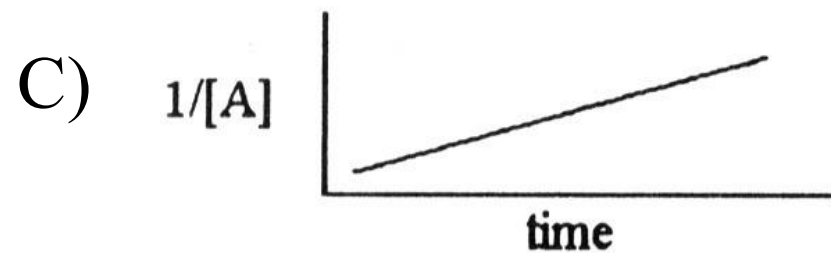
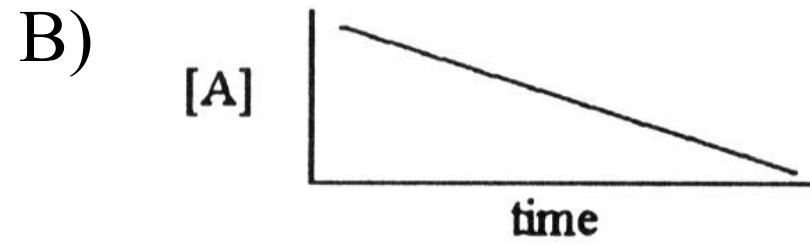
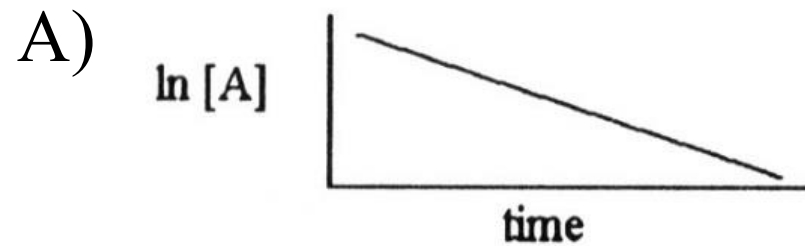
D) 1.2

E) 0.54

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

$$1/0.0065 = 1/0.01 + 100k$$

5. Which one of the following graphs shows the correct relationship between concentration and time for a reaction that is second order in [A]? C



6. The decomposition of N_2O_5 in solution in carbon tetrachloride proceeds via the reaction:



The reaction is first order at 64°C . The rate law for the reaction is rate = C.

A) $k[\text{N}_2\text{O}_5]^2$

B) $k \frac{[\text{NO}_2]^4 [\text{O}_2]}{[\text{N}_2\text{O}_5]^2}$

C) $k[\text{N}_2\text{O}_5]$

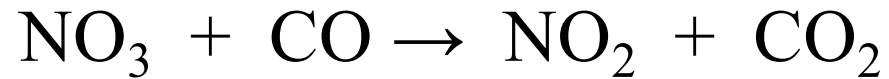
D) $k \frac{[\text{N}_2\text{O}_5]^2}{[\text{NO}_2]^4 [\text{O}_2]}$

E) $2k [\text{N}_2\text{O}_5]$

$$\text{rate} = k[A]^m[B]^n$$

m and n is the reaction order.

7. For the elementary reaction



the molecularity of the reaction is A, and the rate law is
rate=_____

Molecularity: the number of molecules that participate as reactants in an elementary reaction.

- A) 2, $k[\text{NO}_3][\text{CO}]$
- B) 4, $k[\text{NO}_3][\text{CO}][\text{NO}_2][\text{CO}_2]$
- C) 2, $k[\text{NO}_2][\text{CO}_2]$
- D) 2, $k[\text{NO}_3][\text{CO}]/[\text{NO}_2][\text{CO}_2]$
- E) 4, $k[\text{NO}_2][\text{CO}_2]/[\text{NO}_3][\text{CO}]$

8. A catalyst can increase the rate of a reaction D .

A) by changing the value of the frequency factor (A)

B) by increasing the overall activation energy (E_a) of the reaction

C) by lowering the activation energy of the reverse reaction

D) by providing an alternative pathway with a lower activation energy

E) All of these are ways that a catalyst might act to increase the rate of reaction.

9. Which one of the following will change the value of an equilibrium constant? A

A) changing temperature

B) adding other substances that do not react with any of the species involved in the equilibrium

C) varying the initial concentrations of reactants

D) varying the initial concentrations of products

E) changing the volume of the reaction vessel